

Straightening Laws in Algebraic Complexity

Anakin Dey

What is Algebraic Complexity?

Question

How hard is it to compute polynomials?

What is Algebraic Complexity?

Question

How hard is it to compute polynomials in restricted models?

What is Algebraic Complexity?

Question

How hard is it to compute polynomials in restricted models?

Question

Can we find *explicit* polynomials which are hard to compute?

Algebraic Circuits

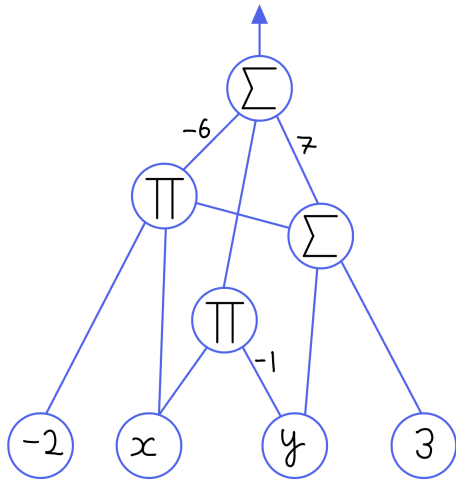
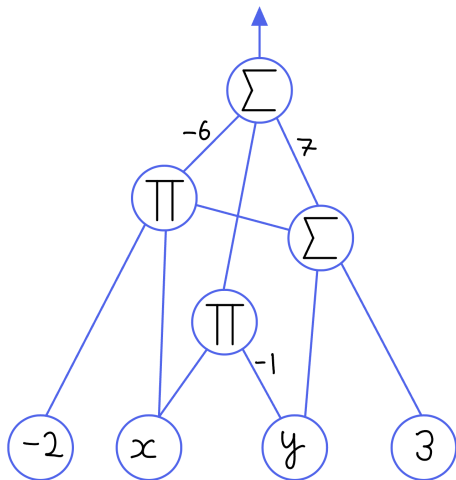


Figure: $-6(-2x \cdot (y + 3)) - xy + 7(y + 3)$

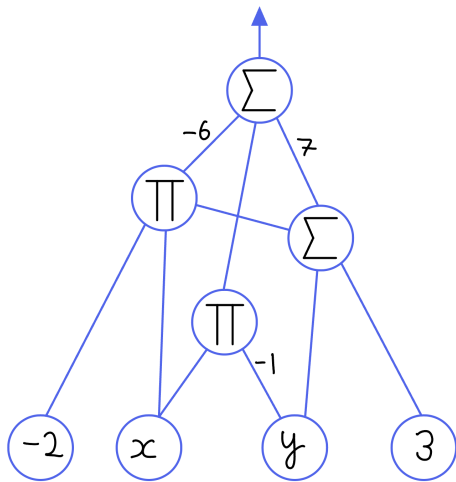
Algebraic Circuits



We have two measures of complexity:

Figure: $-6(-2x \cdot (y + 3)) - xy + 7(y + 3)$

Algebraic Circuits

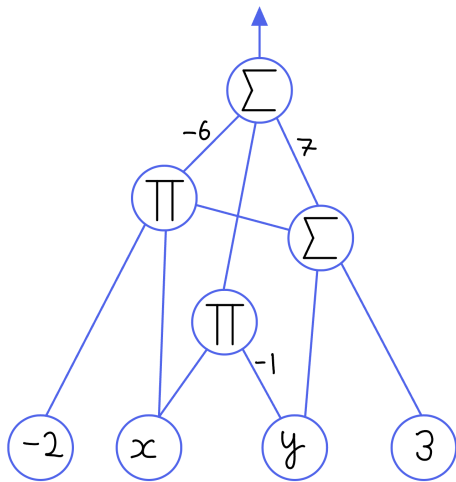


We have two measures of complexity:

- *Size*: The number of edges

Figure: $-6(-2x \cdot (y + 3)) - xy + 7(y + 3)$

Algebraic Circuits



We have two measures of complexity:

- *Size*: The number of edges
- *Depth*: The length of the longest input-output path

Figure: $-6(-2x \cdot (y + 3)) - xy + 7(y + 3)$

Horner's Rule

Let $f(x) = \sum_{i=0}^d a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

Horner's Rule

Let $f(x) = \sum_{i=0}^d a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

Example

Horner's Rule gives a recurrence for efficiently computing $f(x)$:

$$p_0(x) = a_d, \quad p_k(x) = a_{d-k} + x \cdot p_{k-1}(x).$$

Horner's Rule

Let $f(x) = \sum_{i=0}^d a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

Example

Horner's Rule gives a recurrence for efficiently computing $f(x)$:

$$p_0(x) = a_d, \quad p_k(x) = a_{d-k} + x \cdot p_{k-1}(x).$$

- This gives a circuit using exactly d additions and d multiplications.

Horner's Rule

Let $f(x) = \sum_{i=0}^d a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

Example

Horner's Rule gives a recurrence for efficiently computing $f(x)$:

$$p_0(x) = a_d, \quad p_k(x) = a_{d-k} + x \cdot p_{k-1}(x).$$

- This gives a circuit using exactly d additions and d multiplications.
- The number of additions is optimal [Ost54].
- The number of multiplications is optimal [Pan66].

Hard Polynomials Exist

Proposition

Let $\mathbb{F}$ be a field of characteristic zero. Then “most” n -variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

Hard Polynomials Exist

Proposition

Let $\mathbb{F}$ be a field of characteristic zero. Then “most” n -variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

Sketch

- We want the set of polynomials computable by a circuit of size s to be Zariski-dense in $\mathbb{F}[x_1, \dots, x_n]_{\leq d} \simeq \mathbb{A}_F^N$.

Hard Polynomials Exist

Proposition

Let $\mathbb{F}$ be a field of characteristic zero. Then “most” n -variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

Sketch

- We want the set of polynomials computable by a circuit of size s to be Zariski-dense in $\mathbb{F}[x_1, \dots, x_n]_{\leq d} \simeq \mathbb{A}_F^N$.
- For an *unlabeled* circuit of size s

Hard Polynomials Exist

Proposition

Let $\mathbb{F}$ be a field of characteristic zero. Then “most” n -variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

Sketch

- We want the set of polynomials computable by a circuit of size s to be Zariski-dense in $\mathbb{F}[x_1, \dots, x_n]_{\leq d} \simeq \mathbb{A}_{\mathbb{F}}^N$.
- For an *unlabeled* circuit of size s , labeling the edges gives a map

$$\Phi: \mathbb{A}_{\mathbb{F}}^s \rightarrow \mathbb{A}_{\mathbb{F}}^N$$

Hard Polynomials Exist

Proposition

Let $\mathbb{F}$ be a field of characteristic zero. Then “most” n -variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

Sketch

- We want the set of polynomials computable by a circuit of size s to be Zariski-dense in $\mathbb{F}[x_1, \dots, x_n]_{\leq d} \simeq \mathbb{A}_{\mathbb{F}}^N$.
- For an *unlabeled* circuit of size s , labeling the edges gives a map

$$\Phi: \mathbb{A}_{\mathbb{F}}^s \rightarrow \mathbb{A}_{\mathbb{F}}^N$$

- $\dim \operatorname{im} \Phi \leq s$, and there are only finitely many unlabeled circuits.

Hard Polynomials Exist

Proposition

Let $\mathbb{F}$ be a field of characteristic zero. Then “most” n -variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

Sketch

- We want the set of polynomials computable by a circuit of size s to be Zariski-dense in $\mathbb{F}[x_1, \dots, x_n]_{\leq d} \simeq \mathbb{A}_{\mathbb{F}}^N$.
- For an *unlabeled* circuit of size s , labeling the edges gives a map

$$\Phi: \mathbb{A}_{\mathbb{F}}^s \rightarrow \mathbb{A}_{\mathbb{F}}^N$$

- $\dim \operatorname{im} \Phi \leq s$, and there are only finitely many unlabeled circuits.
- Thus, the dimension over all possible unlabeled circuits is at most s .

Hard Polynomials Exist

Proposition

Let $\mathbb{F}$ be a field of characteristic zero. Then “most” n -variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

Sketch

- We want the set of polynomials computable by a circuit of size s to be Zariski-dense in $\mathbb{F}[x_1, \dots, x_n]_{\leq d} \simeq \mathbb{A}_{\mathbb{F}}^N$.
- For an *unlabeled* circuit of size s , labeling the edges gives a map

$$\Phi: \mathbb{A}_{\mathbb{F}}^s \rightarrow \mathbb{A}_{\mathbb{F}}^N$$

- $\dim \operatorname{im} \Phi \leq s$, and there are only finitely many unlabeled circuits.
- Thus, the dimension over all possible unlabeled circuits is at most s .
- For the image to be Zariski-dense, we must have that $s \geq N$.

Determinant versus Permanent

Conjecture

Let $\mathbb{F}$ be a field, $\text{char}(\mathbb{F}) \neq 2$, $X = (x_{i,j})$ be $n \times n$ matrix of variables.

Determinant versus Permanent

Conjecture

Let $\mathbb{F}$ be a field, $\text{char}(\mathbb{F}) \neq 2$, $X = (x_{i,j})$ be $n \times n$ matrix of variables.

Suppose that A is an $m \times m$ matrix with entries in $\mathbb{F}[X]_{\leq 1}$ such that

$$\det_m(A) = \text{perm}_n(X).$$

Determinant versus Permanent

Conjecture

Let $\mathbb{F}$ be a field, $\text{char}(\mathbb{F}) \neq 2$, $X = (x_{i,j})$ be $n \times n$ matrix of variables.

Suppose that A is an $m \times m$ matrix with entries in $\mathbb{F}[X]_{\leq 1}$ such that

$$\det_m(A) = \text{perm}_n(X).$$

Then m must be *superpolynomial* in n .

Determinant versus Permanent

Conjecture

Let $\mathbb{F}$ be a field, $\text{char}(\mathbb{F}) \neq 2$, $X = (x_{i,j})$ be $n \times n$ matrix of variables.

Suppose that A is an $m \times m$ matrix with entries in $\mathbb{F}[X]_{\leq 1}$ such that

$$\det_m(A) = \text{perm}_n(X).$$

Then m must be *superpolynomial* in n .

- We know of $\text{poly}(m)$ -sized circuits for $\det_m$
- It turns out the conjecture is equivalent to saying that perm_n requires superpolynomial-sized circuits

Border Complexity

One relaxation in algebraic complexity is to *approximate* a polynomial.

Border Complexity

One relaxation in algebraic complexity is to *approximate* a polynomial.

Instead of working over a field $\mathbb{F}$

Border Complexity

One relaxation in algebraic complexity is to *approximate* a polynomial.

Instead of working over a field $\mathbb{F}$, we take a parameter ε , allow our coefficients to be in $\mathbb{F}(\varepsilon)$, and let ε tend to 0.

Border Complexity

One relaxation in algebraic complexity is to *approximate* a polynomial.

Instead of working over a field $\mathbb{F}$, we take a parameter ε , allow our coefficients to be in $\mathbb{F}(\varepsilon)$, and let ε tend to 0.

Definition

We say a circuit *border computes* $g(\overline{x})$ over $\mathbb{F}(\varepsilon)$ if it computes

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\overline{x}], \quad q \geq 1.$$

Border Complexity

One relaxation in algebraic complexity is to *approximate* a polynomial.

Instead of working over a field $\mathbb{F}$, we take a parameter ε , allow our coefficients to be in $\mathbb{F}(\varepsilon)$, and let ε tend to 0.

Definition

We say a circuit *border computes* $g(\overline{x})$ over $\mathbb{F}(\varepsilon)$ if it computes

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\overline{x}], \quad q \geq 1.$$

We **cannot** set $\varepsilon = 0$ since we may divide by ε .

What does Border Computation Get Us?

Question

What is the power of border computation?

What does Border Computation Get Us?

Question

What is the power of border computation?

Is it strictly necessary?

What does Border Computation Get Us?

Question

What is the power of border computation?

Is it strictly necessary? Can we prove something in the border setting and *deborder* the result to get an exact computation?

Border Waring Rank

Definition

The *Waring Rank*, $\text{WR}(f)$, of a degree d form $f(\bar{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\bar{x}), \dots, \ell_r(\bar{x})$ such that

$$f(\bar{x}) = \ell_1(\bar{x})^d + \dots + \ell_r(\bar{x})^d.$$

Border Waring Rank

Definition

The *Waring Rank*, $\text{WR}(f)$, of a degree d form $f(\bar{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\bar{x}), \dots, \ell_r(\bar{x})$ such that

$$f(\bar{x}) = \ell_1(\bar{x})^d + \dots + \ell_r(\bar{x})^d.$$

Example ([DL25])

We have that $\text{WR}(x^{d-1}y) = d$ [Old40].

Border Waring Rank

Definition

The *Waring Rank*, $\text{WR}(f)$, of a degree d form $f(\bar{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\bar{x}), \dots, \ell_r(\bar{x})$ such that

$$f(\bar{x}) = \ell_1(\bar{x})^d + \dots + \ell_r(\bar{x})^d.$$

Example ([DL25])

We have that $\text{WR}(x^{d-1}y) = d$ [Old40]. However,

$$x^{d-1}y = \lim_{\varepsilon \rightarrow 0} \frac{1}{d\varepsilon} ((x + \varepsilon y)^d - x^d).$$

Border Waring Rank

Definition

The *Waring Rank*, $\text{WR}(f)$, of a degree d form $f(\bar{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\bar{x}), \dots, \ell_r(\bar{x})$ such that

$$f(\bar{x}) = \ell_1(\bar{x})^d + \dots + \ell_r(\bar{x})^d.$$

Example ([DL25])

We have that $\text{WR}(x^{d-1}y) = d$ [Old40]. However,

$$x^{d-1}y = \lim_{\varepsilon \rightarrow 0} \frac{1}{d\varepsilon} ((x + \varepsilon y)^d - x^d).$$

Thus, the *Border Waring Rank*, $\overline{\text{WR}}(f)$, is equal to 2.

Border Waring Rank

Definition

The *Waring Rank*, $\text{WR}(f)$, of a degree d form $f(\bar{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\bar{x}), \dots, \ell_r(\bar{x})$ such that

$$f(\bar{x}) = \ell_1(\bar{x})^d + \dots + \ell_r(\bar{x})^d.$$

Example ([DL25])

We have that $\text{WR}(x^{d-1}y) = d$ [Old40]. However,

$$x^{d-1}y = \lim_{\varepsilon \rightarrow 0} \frac{1}{d\varepsilon} ((x + \varepsilon y)^d - x^d).$$

Thus, the *Border Waring Rank*, $\overline{\text{WR}}(f)$, is equal to 2.

$\implies \{ f(x, y) \in \mathbb{C}[x, y]_{\leq d} \mid \text{WR}(f) = 2 \}$ is *not* Zariski-closed.

Circuit Complexity for Ideals

In algebraic complexity, we are interested in characterizing the circuit complexity of *families* of polynomials.

Circuit Complexity for Ideals

In algebraic complexity, we are interested in characterizing the circuit complexity of *families* of polynomials.

Question

Suppose $f \in \langle g_1, \dots, g_k \rangle$.

Circuit Complexity for Ideals

In algebraic complexity, we are interested in characterizing the circuit complexity of *families* of polynomials.

Question

Suppose $f \in \langle g_1, \dots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \dots, g_k$?

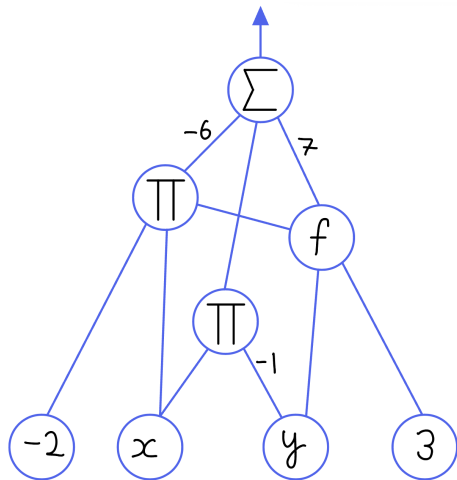
Algebraic Circuits with Oracles

Algebraic Circuits with Oracles

Question

How can we compare the *relative* complexity of two polynomials?

Algebraic Circuits with Oracles



Question

How can we compare the *relative* complexity of two polynomials?

We can allow a fixed polynomial $f(\bar{x})$ to be computed with unit cost by allowing it to function as its own gate.

Figure: $-6(-2x \cdot f(y, 3)) - xy + 7f(y, 3)$

Principal Ideals

Question

Suppose $0 \neq f \in \langle g_1, \dots, g_k \rangle$. Can we compute any of the generators g_i efficiently with an f -oracle circuit?

Principal Ideals

Question

Suppose $0 \neq f \in \langle g_1, \dots, g_k \rangle$. Can we compute any of the generators g_i efficiently with an f -oracle circuit?

Question

Suppose $0 \neq f \in \langle g \rangle$. Does g have a small f -oracle circuit?

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If $f \in \langle g \rangle$, $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$.

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If $f \in \langle g \rangle$, $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$.

Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0, g can be *border computed* by a circuit of size $\text{poly}(\text{size}(f), \deg(f))$.

Principal Ideals

Conjecture ([Bür00, Conjecture 8.3])

If $f \in \langle g \rangle$, $\text{size}(g) \leq \text{poly}(\text{size}(f), \deg(f))$.

Theorem ([Bür04, Theorem 1.3])

Over fields of characteristic 0, g can be *border computed* by a circuit of size $\text{poly}(\text{size}(f), \deg(f))$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

What about Two Generators?

Consider an ideal with two generators $\langle g_1(\overline{x}), g_2(\overline{x}) \rangle$.

What about Two Generators?

Consider an ideal with two generators $\langle g_1(\bar{x}), g_2(\bar{x}) \rangle$.

Say $0 \neq f(\bar{x}) \in \langle g_1(\bar{x}), g_2(\bar{x}) \rangle$:

What about Two Generators?

Consider an ideal with two generators $\langle g_1(\bar{x}), g_2(\bar{x}) \rangle$.

Say $0 \neq f(\bar{x}) \in \langle g_1(\bar{x}), g_2(\bar{x}) \rangle$:

$$f(\bar{x}) = h_1(\bar{x})g_1(\bar{x}) + h_2(\bar{x})g_2(\bar{x}).$$

What about Two Generators?

Consider an ideal with two generators $\langle g_1(\bar{x}), g_2(\bar{x}) \rangle$.

Say $0 \neq f(\bar{x}) \in \langle g_1(\bar{x}), g_2(\bar{x}) \rangle$:

$$f(\bar{x}) = h_1(\bar{x})g_1(\bar{x}) + h_2(\bar{x})g_2(\bar{x}).$$

Question

If $f(\bar{x})$ has a small circuit, do $g_1(\bar{x})$ or $g_2(\bar{x})$ have a small f -oracle circuit?

What about Two Generators?

Consider an ideal with two generators $\langle g_1(\bar{x}), g_2(\bar{x}) \rangle$.

Say $0 \neq f(\bar{x}) \in \langle g_1(\bar{x}), g_2(\bar{x}) \rangle$:

$$f(\bar{x}) = h_1(\bar{x})g_1(\bar{x}) + h_2(\bar{x})g_2(\bar{x}).$$

Question

If $f(\bar{x})$ has a small circuit, do $g_1(\bar{x})$ or $g_2(\bar{x})$ have a small f -oracle circuit?

This is open for *general* polynomials $g_1(\bar{x}), g_2(\bar{x})$ [Gro20].

What about Two Generators?

Consider an ideal with two generators $\langle g_1(\bar{x}), g_2(\bar{x}) \rangle$.

Say $0 \neq f(\bar{x}) \in \langle g_1(\bar{x}), g_2(\bar{x}) \rangle$:

$$f(\bar{x}) = h_1(\bar{x})g_1(\bar{x}) + h_2(\bar{x})g_2(\bar{x}).$$

Question

If $f(\bar{x})$ has a small circuit, do $g_1(\bar{x})$ or $g_2(\bar{x})$ have a small f -oracle circuit?

This is open for *general* polynomials $g_1(\bar{x}), g_2(\bar{x})$ [Gro20].

This is open even if we ask that $\{g_1, g_2\}$ are a Gröbner basis or form a regular sequence.

Weird Statements Can Arise

Question

Suppose $0 \neq f \in \langle g_1, \dots, g_k \rangle$. Can we compute any of the generators g_i efficiently with an f -oracle circuit?

Weird Statements Can Arise

Question

Suppose $0 \neq f \in \langle g_1, \dots, g_k \rangle$. Can we compute any of the generators g_i efficiently with an f -oracle circuit?

Example

- Let $I := \langle \text{perm}_n, \det_n - \text{perm}_n \rangle$.

Weird Statements Can Arise

Question

Suppose $0 \neq f \in \langle g_1, \dots, g_k \rangle$. Can we compute any of the generators g_i efficiently with an f -oracle circuit?

Example

- Let $I := \langle \text{perm}_n, \det_n - \text{perm}_n \rangle$.
- It is conjectured that perm_n , and thus $\det_n - \text{perm}_n$, requires superpolynomial circuits.

Weird Statements Can Arise

Question

Suppose $0 \neq f \in \langle g_1, \dots, g_k \rangle$. Can we compute any of the generators g_i efficiently with an f -oracle circuit?

Example

- Let $I := \langle \text{perm}_n, \det_n - \text{perm}_n \rangle$.
- It is conjectured that perm_n , and thus $\det_n - \text{perm}_n$, requires superpolynomial circuits.
- However, $\det_n = \text{perm}_n + (\det_n - \text{perm}_n) \in I$ is known to have a circuit of size $\text{poly}(n)$ computing it.

Weird Statements Can Arise

Question

Suppose $0 \neq f \in \langle g_1, \dots, g_k \rangle$. Can we compute any of the generators g_i efficiently with an f -oracle circuit?

Example

- Let $I := \langle \text{perm}_n, \det_n - \text{perm}_n \rangle$.
- It is conjectured that perm_n , and thus $\det_n - \text{perm}_n$, requires superpolynomial circuits.
- However, $\det_n = \text{perm}_n + (\det_n - \text{perm}_n) \in I$ is known to have a circuit of size $\text{poly}(n)$ computing it.
- Thus, our question is asking us to resolve $\det_n$ versus perm_n .

Adding $\det_n$ to the set of generators remedies this issue.

Determinantal Ideals

Questions about oracle circuit complexity are open for general ideals with more than one generator.

Determinantal Ideals

Questions about oracle circuit complexity are open for general ideals with more than one generator. We can instead ask about ideals whose generators have *additional structure*.

Determinantal Ideals

Questions about oracle circuit complexity are open for general ideals with more than one generator. We can instead ask about ideals whose generators have *additional structure*.

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables.

Determinantal Ideals

Questions about oracle circuit complexity are open for general ideals with more than one generator. We can instead ask about ideals whose generators have *additional structure*.

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Prior Closure Results in Determinantal Ideals

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Prior Closure Results in Determinantal Ideals

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial.

Prior Closure Results in Determinantal Ideals

Example

Consider an $n \times m$ matrix $X = (x_{i,j})$ of variables. Let $I_{n,m,r}^{\det}$ be the *determinantal ideal* generated by the $r \times r$ minors of X .

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = \text{poly}(n)$.

What does the Oracle Give Us?

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = \text{poly}(n)$.

What does the Oracle Give Us?

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = \text{poly}(n)$.

- There exist polynomial size circuits for computing $\det(X)$.

What does the Oracle Give Us?

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = \text{poly}(n)$.

- There exist polynomial size circuits for computing $\det(X)$.
- However, these circuits are *not* constant-depth.

What does the Oracle Give Us?

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. There is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant, $t = \text{poly}(n)$.

- There exist polynomial size circuits for computing $\det(X)$.
- However, these circuits are *not* constant-depth.
- The oracle allows *constant depth* computation of the determinant.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *border computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $O(n^2 m^2)$ that *border computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

Question

Can we *deborder* this result, that is can we remove this ε approximation?

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(X)$ in a standard basis indexed by Young tableau

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(X)$ in a standard basis indexed by Young tableau, and leverage the combinatorics to talk about specific terms.

Closure Results in Determinantal Ideals

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

Main Tools:

- We use *Straightening Laws* from Invariant Theory to express $f(X)$ in a standard basis indexed by Young tableau, and leverage the combinatorics to talk about specific terms.
- To get a circuit for a specific basis term, we use *Interpolation* as well as a novel application of the *Isolation Lemma*.

Dealing with Cancellations

Dealing with Cancellations

- We saw earlier that we have to be careful of cancellations when working with ideals.

Dealing with Cancellations

- We saw earlier that we have to be careful of cancellations when working with ideals.
- Working with the monomial basis is hard since determinants have exponentially many terms.

Dealing with Cancellations

- We saw earlier that we have to be careful of cancellations when working with ideals.
- Working with the monomial basis is hard since determinants have exponentially many terms.
- Thus, we will work with a different basis for $I_{n,m,r}^{\det}$ which is described combinatorially with Young tableau.

A Natural Generating Set for Determinantal Ideals

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$.

A Natural Generating Set for Determinantal Ideals

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$. Consider the *bitableau* $(S \mid T)$ of shape σ

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 & & \\ \hline 4 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 & & \\ \hline 3 & & & \\ \hline \end{array} \right).$$

A Natural Generating Set for Determinantal Ideals

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$. Consider the *bitableau* $(S \mid T)$ of shape σ

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 & & \\ \hline 4 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 & & \\ \hline 3 & & & \\ \hline \end{array} \right).$$

Then the *bideterminant* $(S \mid T)(X)$ is given by

$$(S \mid T)(X) = \begin{pmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{pmatrix} \begin{pmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{pmatrix} (x_{4,3}).$$

A Natural Generating Set for Determinantal Ideals

Lemma

A degree d monomial $\prod_{i=1}^d x_{r_i, c_i}$ is given by a bideterminant:

A Natural Generating Set for Determinantal Ideals

Lemma

A degree d monomial $\prod_{i=1}^d x_{r_i, c_i}$ is given by a bideterminant:

$$\prod_{i=1}^d x_{r_i, c_i} = \left(\begin{array}{c|c} \boxed{r_1} & \boxed{c_1} \\ \boxed{r_2} & \boxed{c_2} \\ \boxed{\vdots} & \boxed{\vdots} \\ \boxed{r_d} & \boxed{c_d} \end{array} \right) (X).$$

A Natural Generating Set for Determinantal Ideals

Lemma

A degree d monomial $\prod_{i=1}^d x_{r_i, c_i}$ is given by a bideterminant:

$$\prod_{i=1}^d x_{r_i, c_i} = \left(\begin{array}{c|c} \boxed{r_1} & \boxed{c_1} \\ \boxed{r_2} & \boxed{c_2} \\ \boxed{\vdots} & \boxed{\vdots} \\ \boxed{r_d} & \boxed{c_d} \end{array} \right) (X).$$

Thus, the bideterminants span $\mathbb{F}[X]$.

A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ .

A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ . Then $(S \mid T)(X)$ can be uniquely expressed as an integral linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B} (A \mid B)(X), \quad c_{A,B} \in \mathbb{Z}$$

where A and B are *conjugate semistandard*: strictly increasing along rows, weakly increasing down columns.

A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ . Then $(S \mid T)(X)$ can be uniquely expressed as an integral linear combination

$$(S \mid T)(X) = \sum_{(A \mid B)} c_{A,B} (A \mid B)(X), \quad c_{A,B} \in \mathbb{Z}$$

where A and B are *conjugate semistandard*: strictly increasing along rows, weakly increasing down columns.

Furthermore, if $(A \mid B)$ has shape τ , then $\tau \geq \sigma$ (*dominance order*).

A Natural Generating Set for Determinantal Ideals

Since $I_{n,m,r}^{\det}$ is generated by the $r \times r$ minors, the bideterminants in our basis should have at least one determinant of length at least r .

A Natural Generating Set for Determinantal Ideals

Since $I_{n,m,r}^{\det}$ is generated by the $r \times r$ minors, the bideterminants in our basis should have at least one determinant of length at least r .

Proposition ([BC03, Corollary 1.7])

The ideal $I_{n,m,r}^{\det}$ has as basis the conjugate semistandard bideterminants of shape σ whose first row has length at least r .

Shuffle Products

$$\left(\begin{array}{|c|c|c|c|c|c|} \hline i_1 & \cdots & i_p & i_{p+1} & \cdots & i_s \\ \hline l_1 & \cdots & l_q & l_{q+1} & \cdots & l_t \\ \hline \end{array} \quad \bigg| \quad \begin{array}{|c|c|c|} \hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline \end{array} \right)$$

$$= \sum_{\sigma} \operatorname{sgn}(\sigma) \left(\begin{array}{|c|c|c|c|c|c|} \hline \sigma(i_1) & \cdots & \sigma(i_p) & i_{p+1} & \cdots & i_s \\ \hline \sigma(l_1) & \cdots & \sigma(l_q) & l_{q+1} & \cdots & l_t \\ \hline \end{array} \quad \bigg| \quad \begin{array}{|c|c|c|} \hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline \end{array} \right)$$

Where the summation is over all permutations of $\{i_1, \dots, i_p, l_1, \dots, l_q\}$ such that $\sigma(i_1) < \dots < \sigma(i_p)$ and $\sigma(l_1) < \dots < \sigma(l_q)$.

Laplace Expansion

$$\left(\begin{array}{|c|c|c|c|c|c|} \hline i_1 & \dots & i_p & i_{p+1} & \dots & i_s \\ \hline l_1 & \dots & l_q & l_{q+1} & \dots & l_t \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline j_1 & \dots & j_s \\ \hline m_1 & \dots & m_t \\ \hline \end{array} \right)$$

$$= \sum_{\tau, \mu} \text{sgn}(\tau) \text{sgn}(\mu) \left(\begin{array}{|c|c|c|c|c|c|} \hline i_1 & \dots & i_p & l_1 & \dots & l_q \\ \hline i_{p+1} & \dots & i_s & & & \\ \hline l_{q+1} & \dots & l_t & & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|c|c|c|} \hline \tau(j_1) & \dots & \tau(j_p) & \mu(m_1) & \dots & \mu(m_t) \\ \hline \tau(j_{p+1}) & \dots & \tau(j_s) & & & \\ \hline \mu(m_{q+1}) & \dots & \mu(m_t) & & & \\ \hline \end{array} \right)$$

Where the summation is over all permutations of $\{j_1, \dots, j_p, j_{p+1}, \dots, j_s\}$ such that $\tau(j_1) < \dots < \tau(j_p)$ and $\tau(j_{p+1}) < \dots < \tau(j_s)$ and similar for μ .

Example Straightening

Consider the following nonstandard tableau

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right)$$

Example Straightening

Shuffle:

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) + \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right)$$

Example Straightening

Shuffle:

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) + \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right)$$

Laplace:

$$\begin{aligned} \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) &= \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 1 & 2 \\ \hline 3 & & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 3 & 1 & 2 \\ \hline 1 & & \\ \hline \end{array} \right) \\ &= - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline \end{array} \right) \end{aligned}$$

Example Straightening

Shuffle:

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) + \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right)$$

Laplace:

$$\begin{aligned} \left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) &= \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 1 & 2 \\ \hline 3 & & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 3 & 1 & 2 \\ \hline 1 & & \\ \hline \end{array} \right) \\ &= - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline \end{array} \right) \end{aligned}$$

We notice that our unstraightened rows appear in the first expression so that we can solve for it.

Example Straightening

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline \end{array} \right)$$

Example Straightening

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline \end{array} \right)$$

Thus,

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline 2 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline 1 & & \\ \hline \end{array} \right)$$

Example Straightening

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline \end{array} \right)$$

Thus,

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline 2 & \\ \hline \end{array} \middle| \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 & \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & & \\ \hline 2 & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 1 & & \\ \hline 1 & & \\ \hline \end{array} \right)$$

All terms that appear have the same content and they are either the same shape with a lexicographically smaller filling or they have a longer shape.

Main Theorem

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. Then there is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant for some $t = \text{poly}(n)$.

Main Theorem

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I_{n,n,n/2}^{\det}$ be a nonzero polynomial. Then there is a constant depth f -oracle circuit of size $\text{poly}(n)$ that computes the $t \times t$ determinant for some $t = \text{poly}(n)$.

Theorem ([DG25, Theorem 1.5])

Let $f \in I_{n,m,r}^{\det}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O(r^{1/3})$.

Main Theorem

Theorem

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then, there exists a depth-three f -oracle circuit of size $O(n^2 m^2 d^3 (n^3 + m^3))$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

Main Theorem

Theorem

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then, there exists a depth-three f -oracle circuit of size $O(n^2 m^2 d^3 (n^3 + m^3))$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

Definition (Canonical Bideterminant [dCEP80])

Let $\sigma = (4, 2, 1)$:

$$(K_\sigma \mid K_\sigma)(X) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} \middle| \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} \right) (X)$$

= product of upper-left minors.

Proof Idea

Since we are starting with $f(X) \in I_{n,m,r}^{\det} \subseteq \mathbb{F}[X]$

Proof Idea

Since we are starting with $f(X) \in I_{n,m,r}^{\det} \subseteq \mathbb{F}[X]$, we can express in terms of our standard basis:

$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X).$$

Proof Idea

Since we are starting with $f(X) \in I_{n,m,r}^{\det} \subseteq \mathbb{F}[X]$, we can express in terms of our standard basis:

$$f(X) = \sum_{k \in I} c_k(S_k \mid T_k)(X).$$

Goal: Use $f(X)$ as an oracle to compute a canonical bideterminant.

Substitution

Definition (Substitution Sperator [dCEP80])

For $i < j$, the operator $\text{Sub}_{i \rightarrow j}$ takes a tableau S and returns the tableau S' formed by taking each row of S which has an i but not a j , replacing that i with j , and sorting the row in increasing order.

Substitution

Definition (Substitution Sperator [dCEP80])

For $i < j$, the operator $\text{Sub}_{i \rightarrow j}$ takes a tableau S and returns the tableau S' formed by taking each row of S which has an i but not a j , replacing that i with j , and sorting the row in increasing order.

$$\text{Sub}_{2 \rightarrow 4} \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 4 & & \\ \hline 1 & & & \\ \hline \end{array}$$

Substitution

Definition (Substitution Sperator [dCEP80])

For $i < j$, the operator $\text{Sub}_{i \rightarrow j}$ takes a tableau S and returns the tableau S' formed by taking each row of S which has an i but not a j , replacing that i with j , and sorting the row in increasing order.

$$\text{Sub}_{2 \rightarrow 4} \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & & \\ \hline 1 & & & \\ \hline \end{array} = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 4 & & \\ \hline 1 & & & \\ \hline \end{array}$$

Let $h_i^j(S)$ be the number of times i is replaced by j in S .

Substitution

These operators send conjugate semistandard tableau to conjugate semistandard tableau.

Substitution

These operators send conjugate semistandard tableau to conjugate semistandard tableau.

In fact, after enough applications we arrive at *anticanonical* tableau:

Substitution

These operators send conjugate semistandard tableau to conjugate semistandard tableau.

In fact, after enough applications we arrive at *anticanonical* tableau:

Let $\sigma = (4, 2, 1)$ and $n = m = 6$:

$$(\overline{K}_\sigma \mid \overline{K}_\sigma)(X) = \left(\begin{array}{|c|c|c|c|} \hline 3 & 4 & 5 & 6 \\ \hline 5 & 6 & & \\ \hline 6 & & & \\ \hline \end{array} \mid \begin{array}{|c|c|c|c|} \hline 3 & 4 & 5 & 6 \\ \hline 5 & 6 & & \\ \hline 6 & & & \\ \hline \end{array} \right) (X)$$

= product of lower-right minors.

Substitution

These operators send conjugate semistandard tableau to conjugate semistandard tableau.

In fact, after enough applications we arrive at *anticanonical* tableau:

Substitution

These operators send conjugate semistandard tableau to conjugate semistandard tableau.

In fact, after enough applications we arrive at *anticanonical* tableau:

Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n . Then

$$\begin{aligned} & \left(\underset{n-1 \rightarrow n}{\text{Sub}} \circ \underset{n-2 \rightarrow n}{\text{Sub}} \circ \underset{n-2 \rightarrow n-1}{\text{Sub}} \circ \cdots \right. \\ & \quad \left. \circ \underset{2 \rightarrow n}{\text{Sub}} \circ \cdots \circ \underset{2 \rightarrow 3}{\text{Sub}} \circ \underset{1 \rightarrow n}{\text{Sub}} \circ \cdots \circ \underset{1 \rightarrow 2}{\text{Sub}} \right)(S) = \overline{K}_\sigma. \end{aligned}$$

Linear Operators

Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}(\lambda)$, adding λ times row j to row i .

Linear Operators

Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}(\lambda)$, adding λ times row j to row i .

Lemma

Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau.

Linear Operators

Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}(\lambda)$, adding λ times row j to row i .

Lemma

Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau.

$$(S \mid T)(E_{i,j}(\lambda)X)$$

Linear Operators

Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}(\lambda)$, adding λ times row j to row i .

Lemma

Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau.

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X)$$

Linear Operators

Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}(\lambda)$, adding λ times row j to row i .

Lemma

Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau.

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i \rightarrow j}^h(S)} \pm (S' \mid T)(X).$$

For $0 \leq h \leq h_i^j(S) - 1$, let $\mathcal{C}_{i \rightarrow j}^h(S)$ be the set of tableaux of shape σ obtained by changing i to j at exactly h rows of S which contain i but not j and reordering those rows to be increasing.

Applying all Substitution Operators

Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n . Then

$$\begin{aligned} & \left(\underset{n-1 \rightarrow n}{\text{Sub}} \circ \underset{n-2 \rightarrow n}{\text{Sub}} \circ \underset{n-2 \rightarrow n-1}{\text{Sub}} \circ \cdots \right. \\ & \quad \left. \circ \underset{2 \rightarrow n}{\text{Sub}} \circ \cdots \circ \underset{2 \rightarrow 3}{\text{Sub}} \circ \underset{1 \rightarrow n}{\text{Sub}} \circ \cdots \circ \underset{1 \rightarrow 2}{\text{Sub}} \right)(S) = \overline{K}_\sigma. \end{aligned}$$

Applying all Substitution Operators

Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n . Then

$$\begin{aligned} & \left(\underset{n-1 \rightarrow n}{\text{Sub}} \circ \underset{n-2 \rightarrow n}{\text{Sub}} \circ \underset{n-2 \rightarrow n-1}{\text{Sub}} \circ \cdots \right. \\ & \quad \left. \circ \underset{2 \rightarrow n}{\text{Sub}} \circ \cdots \circ \underset{2 \rightarrow 3}{\text{Sub}} \circ \underset{1 \rightarrow n}{\text{Sub}} \circ \cdots \circ \underset{1 \rightarrow 2}{\text{Sub}} \right)(S) = \overline{K}_\sigma. \end{aligned}$$

$$\begin{aligned} M = & E_{1,2}(\lambda_{1,2}) \cdots E_{1,n}(\lambda_{1,n}) E_{2,3}(\lambda_{2,3}) \cdots E_{2,n}(\lambda_{2,n}) \\ & \cdots E_{n-2,n-1}(\lambda_{n-2,n-1}) E_{n-2,n}(\lambda_{n-2,n}) E_{n-1,n}(\lambda_{n-1,n}) \end{aligned}$$

$$\begin{aligned} N = & E_{m-1,m}(\xi_{m-1,m})^\top E_{m-2,m}(\xi_{m-2,m})^\top E_{m-2,m-1}(\xi_{m-2,m-1})^\top \\ & \cdots E_{2,m}(\xi_{2,m})^\top \cdots E_{2,3}(\xi_{2,3})^\top E_{1,m}(\xi_{1,m})^\top \cdots E_{1,2}(\xi_{1,2})^\top. \end{aligned}$$

Applying all Substitution Operators

We have seen now that after applying a sequence of substitution operators, we can send a bitableau $(S \mid T)$ of shape σ to $(\overline{K}_\sigma \mid \overline{K}_\sigma)$.

Applying all Substitution Operators

We have seen now that after applying a sequence of substitution operators, we can send a bitableau $(S \mid T)$ of shape σ to $(\overline{K}_\sigma \mid \overline{K}_\sigma)$.

Applying the analogous linear operators $E_{i,j}(\lambda_{i,j})$ gives the following:

Applying all Substitution Operators

We have seen now that after applying a sequence of substitution operators, we can send a bitableau $(S \mid T)$ of shape σ to $(\overline{K}_\sigma \mid \overline{K}_\sigma)$.

Applying the analogous linear operators $E_{i,j}(\lambda_{i,j})$ gives the following:

Lemma ([DG25, Lemma 3.9])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d .

Applying all Substitution Operators

We have seen now that after applying a sequence of substitution operators, we can send a bitableau $(S \mid T)$ of shape σ to $(\overline{K}_\sigma \mid \overline{K}_\sigma)$.

Applying the analogous linear operators $E_{i,j}(\lambda_{i,j})$ gives the following:

Lemma ([DG25, Lemma 3.9])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MXN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (\overline{K}_{\sigma_k} \mid \overline{K}_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Applying all Substitution Operators

Lemma ([DG25, Lemma 3.9])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MXN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (\bar{K}_{\sigma_k} \mid \bar{K}_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Applying all Substitution Operators

Lemma ([DG25, Lemma 3.9])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MXN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (\bar{K}_{\sigma_k} \mid \bar{K}_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Let J_n be the $n \times n$ matrix with 1's along the anti-diagonal.

Applying all Substitution Operators

Lemma ([DG25, Lemma 3.9])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MXN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (\bar{K}_{\sigma_k} \mid \bar{K}_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Let J_n be the $n \times n$ matrix with 1's along the anti-diagonal.

Lemma

$$(\bar{K}_\sigma \mid \bar{K}_\sigma)(J_n X J_m) = \pm (K_\sigma \mid K_\sigma)(X).$$

Applying all Substitution Operators

Lemma ([DG25, Corollary 3.10])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MJ_nXJ_mN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Applying all Substitution Operators

Lemma ([DG25, Corollary 3.10])

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Then $f(MXN)$ can be expressed as a sum

$$f(MJ_nXJ_mN) = \sum_{k \in A} \hat{c}_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell \in B} \hat{c}_\ell \Lambda^{\bar{e}_\ell} \Xi^{\bar{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

Distinguishing Terms

$$D = \text{diag}(y_1 v, y_2 v^2, \dots, y_n v^n) \in \mathbb{F}[Y, v]^{n \times n}$$

$$D' = \text{diag}(z_1 v, z_2 v^2, \dots, z_m v^m) \in \mathbb{F}[Z, v]^{m \times m}$$

Distinguishing Terms

$$D = \text{diag}(y_1 v, y_2 v^2, \dots, y_n v^n) \in \mathbb{F}[Y, v]^{n \times n}$$
$$D' = \text{diag}(z_1 v, z_2 v^2, \dots, z_m v^m) \in \mathbb{F}[Z, v]^{m \times m}$$

Lemma

Let $(S \mid T)(X)$ be a bideterminant. Then

$$(S \mid T)(DXD') = Y^{\bar{s}} Z^{\bar{t}} v^{|S|+|T|} \cdot (S \mid T)(X).$$

- $(\bar{s}, \bar{t})$ is the multidegree of $(S \mid T)(X)$: $\text{mdeg}(x_{i,j}) = e_i \oplus e_j$
- $|S|$ is the sum of the entries in S .

Isolating Terms

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

Isolating Terms

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

We can compute g with a layer of Σ -gates followed by a single f -gate.

Isolating Terms

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

We can compute g with a layer of Σ -gates followed by a single f -gate.

We now *interpolate* to get an f -oracle circuit computing $\text{coeff}_{v^{d_{\min}}}(g)$.

Interpolation

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

Interpolation

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

Lemma (Folklore)

Say g is computed by a size s , depth Δ f -oracle circuit with top Σ -gate.

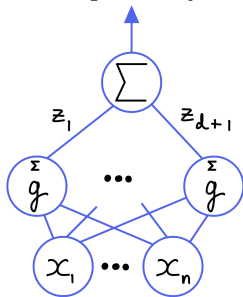
Interpolation

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

Lemma (Folklore)

Say g is computed by a size s , depth Δ f -oracle circuit with top Σ -gate.



Then, we can compute $\text{coeff}_{t^i}(g)$ by a size $O(ds)$, depth Δ f -oracle circuit.

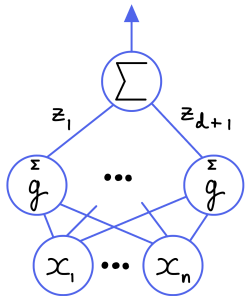
Interpolation

Definition

Consider a degree d polynomial $g(\bar{x}, t) = \sum_{i=0}^d \text{coeff}_{t^i}(g)t^i$.

Lemma (Folklore)

Say g is computed by a size s , depth Δ f -oracle circuit with top Σ -gate.



Then, we can compute $\text{coeff}_{t^i}(g)$ by a size $O(ds)$, depth Δ f -oracle circuit.

Proof

Express *Lagrange interpolation* as a circuit.

Where Does Isolation Arise?

Where Does Isolation Arise?

We now have an f -oracle circuit to compute

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) \in \mathbb{F}[\Lambda, \Xi, Y, Z][X].$$

Where Does Isolation Arise?

We now have an f -oracle circuit to compute

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) \in \mathbb{F}[\Lambda, \Xi, Y, Z][X].$$

We want an f -oracle circuit for a *single* $k \in I$.

Where Does Isolation Arise?

We now have an f -oracle circuit to compute

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) \in \mathbb{F}[\Lambda, \Xi, Y, Z][X].$$

We want an f -oracle circuit for a *single* $k \in I$.

Idea: Interpolate with respect to $\Lambda \cup \Xi \cup Y \cup Z$.

Where Does Isolation Arise?

We now have an f -oracle circuit to compute

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) \in \mathbb{F}[\Lambda, \Xi, Y, Z][X].$$

We want an f -oracle circuit for a *single* $k \in I$.

Idea: Interpolate with respect to $\Lambda \cup \Xi \cup Y \cup Z$.

Problem: Interpolating with respect to multiple variables is *expensive*.

Issues with Interpolation

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Issues with Interpolation

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Recall: Prior work [AF22] gave ideal closure results in the border setting.

If a circuit border computes $g(\bar{x})$, then it computes

$$g(\bar{x}) + \varepsilon^q \tilde{g}(\bar{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\bar{x}], \quad q \geq 1.$$

Issues with Interpolation

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Recall: Prior work [AF22] gave ideal closure results in the border setting.

If a circuit border computes $g(\bar{x})$, then it computes

$$g(\bar{x}) + \varepsilon^q \tilde{g}(\bar{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\bar{x}], \quad q \geq 1.$$

Idea: Interpolate with respect to ε .

Issues with Interpolation

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Recall: Prior work [AF22] gave ideal closure results in the border setting.

If a circuit border computes $g(\bar{x})$, then it computes

$$g(\bar{x}) + \varepsilon^q \tilde{g}(\bar{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\bar{x}], \quad q \geq 1.$$

Idea: Interpolate with respect to ε .

Problem: q can be *exponentially large*

Issues with Interpolation

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Recall: Prior work [AF22] gave ideal closure results in the border setting.

If a circuit border computes $g(\bar{x})$, then it computes

$$g(\bar{x}) + \varepsilon^q \tilde{g}(\bar{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\bar{x}], \quad q \geq 1.$$

Idea: Interpolate with respect to ε .

Problem: q can be *exponentially large*

$\implies$ Interpolation gives *large circuit*.

Idea: Kronecker Substitution

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Idea: Kronecker Substitution

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Lemma

Suppose that $\deg(g) \leq d$

Idea: Kronecker Substitution

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Lemma

Suppose that $\deg(g) \leq d$, then the *Kronecker substitution*

$$x_i \mapsto w^{d^i}$$

maps distinct monomials to distinct monomials.

Idea: Kronecker Substitution

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Lemma

Suppose that $\deg(g) \leq d$, then the *Kronecker substitution*

$$x_i \mapsto w^{d^i}$$

maps distinct monomials to distinct monomials.

Problem: The resulting polynomial has *large degree*.

Idea: Kronecker Substitution

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Lemma

Suppose that $\deg(g) \leq d$, then the *Kronecker substitution*

$$x_i \mapsto w^{d^i}$$

maps distinct monomials to distinct monomials.

Problem: The resulting polynomial has *large degree*.

$\implies$ Interpolation gives *large circuit*.

Isolation Lemma

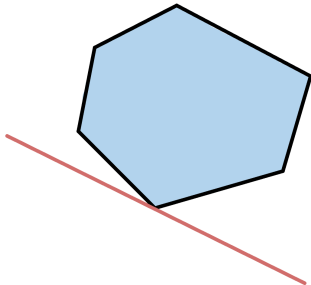
Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Isolation Lemma

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Lemma ([KS01, Lemma 4])

Linear programs with *random* cost functions will have a unique minimum.

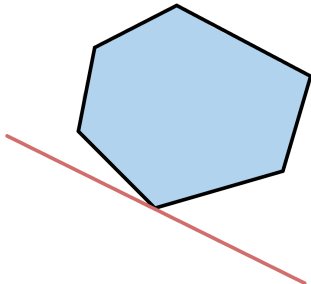


Isolation Lemma

Setup: We want to *isolate* some coefficient $c_{\bar{e}}$ in $g(\bar{x}) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \bar{x}^{\bar{e}}$.

Lemma ([KS01, Lemma 4])

Linear programs with *random* cost functions will have a unique minimum.



Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

Isolation Lemma

Lemma ([KS01, Lemma 4])

Let $\mathcal{L}$ be any collection of distinct linear forms in variables $z_1, \dots, z_\ell$ with coefficients in the range $\{0, \dots, K\}$ for some integer $K \in \mathbb{Z}_{\geq 0}$. Let $\varepsilon > 0$.

Isolation Lemma

Lemma ([KS01, Lemma 4])

Let $\mathcal{L}$ be any collection of distinct linear forms in variables $z_1, \dots, z_\ell$ with coefficients in the range $\{0, \dots, K\}$ for some integer $K \in \mathbb{Z}_{\geq 0}$. Let $\varepsilon > 0$.

Let $z_1, \dots, z_\ell$ be independently and uniformly chosen from $\{0, \dots, M\}$ at random, where $M \geq K\ell/\varepsilon$.

Isolation Lemma

Lemma ([KS01, Lemma 4])

Let $\mathcal{L}$ be any collection of distinct linear forms in variables $z_1, \dots, z_\ell$ with coefficients in the range $\{0, \dots, K\}$ for some integer $K \in \mathbb{Z}_{\geq 0}$. Let $\varepsilon > 0$.

Let $z_1, \dots, z_\ell$ be independently and uniformly chosen from $\{0, \dots, M\}$ at random, where $M \geq K\ell/\varepsilon$.

Then, with probability at least $1 - \varepsilon$, there is a unique form in $\mathcal{L}$ of minimum value at $z_1, \dots, z_\ell$.

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}$$

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

The Isolation Lemma shows that the $z_1, \dots, z_\ell$ can be chosen to be small

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

The Isolation Lemma shows that the $z_1, \dots, z_\ell$ can be chosen to be small
 $\implies \exists$ unique lowest $\deg_w$ -term

Isolating Monomials

Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \dots, x_\ell)$ such that the individual degree of each x_i in g is at most K :

$$g(x_1, \dots, x_\ell) = \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} x_1^{e_1} \cdots x_\ell^{e_\ell}.$$

Randomly choose $z_1, \dots, z_\ell$ and define a morphism

$$x_i \mapsto w^{z_i}, \quad g \mapsto \sum_{\bar{e} \in \mathcal{L}} c_{\bar{e}} \cdot w^{\sum_{i=1}^{\ell} e_i z_i}.$$

The Isolation Lemma shows that the $z_1, \dots, z_\ell$ can be chosen to be small
 $\implies \exists$ unique lowest $\deg_w$ -term and interpolation w.r.t w is small.

Isolating Terms

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

Isolating Terms

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

Compute g with a layer of Σ -gates followed by a single f -gate.

Isolating Terms

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

Compute g with a layer of Σ -gates followed by a single f -gate.

Then *interpolate* to get an f -oracle circuit computing $\text{coeff}_{v^{d_{\min}}}(g)$.

Isolating Terms

Lemma

Let $f(X) \in I_{n,m,r}^{\det}$ be a nonzero polynomial of degree d . Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\text{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

Compute g with a layer of Σ -gates followed by a single f -gate.

Then *interpolate* to get an f -oracle circuit computing $\text{coeff}_{v^{d_{\min}}}(g)$.

Apply the *Isolation Lemma* to get an f -oracle circuit computing some

$$(K_{\sigma_k} \mid K_{\sigma_k})(X).$$

Pfaffian Analogues

Definition

If X is a $2n \times 2n$ skew-symmetric matrix,

Pfaffian Analogues

Definition

If X is a $2n \times 2n$ skew-symmetric matrix, then $\det(X)$ is the perfect square of another polynomial called the *Pfaffian* of X .

Pfaffian Analogues

Definition

If X is a $2n \times 2n$ skew-symmetric matrix, then $\det(X)$ is the perfect square of another polynomial called the *Pfaffian* of X .

Theorem ([DG25, Theorem 1.6])

Let $f \in I_{2n, 2r}^{\text{pfaff}}$ be a nonzero polynomial. Then there is a depth-three f -oracle circuit of size $\text{poly}(n, \deg(f))$ that *exactly computes* the $t \times t$ Pfaffian for some $t = O(r^{1/3})$.

Symmetric Analogue?

Let $x_{i,j}$ be indeterminates for $1 \leq i \leq j \leq n$ and let $x_{j,i} := x_{i,j}$ for $1 \leq j < i \leq n$. Thus, $X = (x_{i,j})_{1 \leq i,j \leq n}$ is a symmetric $n \times n$ matrix.

Symmetric Analogue?

Let $x_{i,j}$ be indeterminates for $1 \leq i \leq j \leq n$ and let $x_{j,i} := x_{i,j}$ for $1 \leq j < i \leq n$. Thus, $X = (x_{i,j})_{1 \leq i,j \leq n}$ is a symmetric $n \times n$ matrix.

Conjecture

Let $I_{n,r}^{\text{sym}} \subseteq \mathbb{F}[X]$ be the ideal generated by the $r \times r$ minors of X . Let $f \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial. Then there exists a constant-depth f -oracle circuit of size $\text{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a symmetric matrix for $t = \text{poly}(n)$.

Partial Progress

Theorem

Let $f(X) \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial of degree d . Then, there exists a depth-three f -oracle circuit of size $O(d^3 n^7)$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

Partial Progress

Theorem

Let $f(X) \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial of degree d . Then, there exists a depth-three f -oracle circuit of size $O(d^3 n^7)$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

Our main roadblock for fully resolving the symmetric case is the reduction from a product of determinants to a single determinant.

Roadblocks for the Permanent

The other big star in algebraic complexity is the *permanent* of a matrix.

Roadblocks for the Permanent

The other big star in algebraic complexity is the *permanent* of a matrix.

Question

Is there an analogue to our results for the ideal of permanents of a matrix?

Roadblocks for the Permanent

The other big star in algebraic complexity is the *permanent* of a matrix.

Question

Is there an analogue to our results for the ideal of permanents of a matrix?

The main roadblock is that we have no analogue of the following result:

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm \lambda^{h_i^j(S)} (\text{Sub}_{i \rightarrow j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i \rightarrow j}^h(S)} \pm (S' \mid T)(X).$$

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80, dCEP82])

Let A be a ring, $H \subseteq A$ is a poset.

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80,dCEP82])

Let A be a ring, $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80,dCEP82])

Let A be a ring, $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Let R be a ring, A an R -algebra, and $H \subseteq A$ a finite poset which generates A as an R -algebra.

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80,dCEP82])

Let A be a ring, $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Let R be a ring, A an R -algebra, and $H \subseteq A$ a finite poset which generates A as an R -algebra. A is an *Algebra with a Straightening Law* if

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80,dCEP82])

Let A be a ring, $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Let R be a ring, A an R -algebra, and $H \subseteq A$ a finite poset which generates A as an R -algebra. A is an *Algebra with a Straightening Law* if

- A is a free R -module with a basis given by the standard monomials

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80,dCEP82])

Let A be a ring, $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Let R be a ring, A an R -algebra, and $H \subseteq A$ a finite poset which generates A as an R -algebra. A is an *Algebra with a Straightening Law* if

- A is a free R -module with a basis given by the standard monomials
- if $\alpha, \beta \in H$ are incomparable, and if

$$\alpha\beta = \sum_{i=1}^m r_i \gamma_{i,1} \cdots \gamma_{i,k_i}$$

where $0 \neq r_i \in R$ and each $\gamma_{i,1} \cdots \gamma_{i,k_i}$ is a standard monomial, we must have that for all $1 \leq i \leq m$ that $\gamma_{i,1} \leq \alpha, \beta$.

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80,dCEP82])

Let A be a ring, $H \subseteq A$ is a poset. A *standard monomial* is a product $a_1 \cdots a_k$ such that $a_i \in H$ and $a_1 \leq \cdots \leq a_k$.

Let R be a ring, A an R -algebra, and $H \subseteq A$ a finite poset which generates A as an R -algebra. A is an *Algebra with a Straightening Law* if

- A is a free R -module with a basis given by the standard monomials
- if $\alpha, \beta \in H$ are incomparable, and if

$$\alpha\beta = \sum_{i=1}^m r_i \gamma_{i,1} \cdots \gamma_{i,k_i}$$

where $0 \neq r_i \in R$ and each $\gamma_{i,1} \cdots \gamma_{i,k_i}$ is a standard monomial, we must have that for all $1 \leq i \leq m$ that $\gamma_{i,1} \leq \alpha, \beta$.

These relations in the second requirement are known as the *straightening relations* or *straightening laws*.

Hankel Ideals

Let $\mathbb{F}$ be a field. Let x_i be indeterminates for $1 \leq i \leq n$. For $1 \leq j \leq n$, let X_j be the following Hankel matrix:

$$\begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-j+1} \\ x_2 & x_3 & x_4 & \cdots & x_{n-j+2} \\ x_3 & x_4 & x_5 & \cdots & x_{n-j+3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_j & x_{j+1} & x_{j+2} & \cdots & x_n \end{pmatrix}.$$

Hankel Ideals

Let $\mathbb{F}$ be a field. Let x_i be indeterminates for $1 \leq i \leq n$. For $1 \leq j \leq n$, let X_j be the following Hankel matrix:

$$\begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-j+1} \\ x_2 & x_3 & x_4 & \cdots & x_{n-j+2} \\ x_3 & x_4 & x_5 & \cdots & x_{n-j+3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_j & x_{j+1} & x_{j+2} & \cdots & x_n \end{pmatrix}.$$

For $1 \leq r \leq \min \{ j, n - j + 1 \}$ let $I_r(X_j)$ be the ideal generated by $r \times r$ minors of X_j .

Hankel Ideals

The ideals $I_r(X_j)$ do not have the structure of an algebra with a straightening law [Con98].

In particular, the product of standard monomials need not be standard.

Hankel Ideals

The ideals $I_r(X_j)$ do not have the structure of an algebra with a straightening law [Con98].

In particular, the product of standard monomials need not be standard.

We still conjecture the following:

Conjecture

Let $I_r(X_j) \subseteq \mathbb{F}[x_1, \dots, x_n]$ be the ideal generated by the $r \times r$ minors of X_j . Let $f \in I_r(X_j)$ be a nonzero polynomial. Then there exists a constant-depth f -oracle circuit of size $\text{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a Hankel matrix for $t = \text{poly}(n)$.

Failure of Prior Strategy

In the general determinantal, skew-symmetric, and symmetric cases we analyzed the behavior of $E_{i,j}(\lambda)$.

Failure of Prior Strategy

In the general determinantal, skew-symmetric, and symmetric cases we analyzed the behavior of $E_{i,j}(\lambda)$.

This no longer works:

$$\begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix} = \begin{pmatrix} x_1 + \lambda x_2 & x_2 + \lambda x_3 & x_3 + \lambda x_4 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix}$$

Thank You!

If you don't know how to answer someone's question, you should compliment them by saying it's a good question.

— Noga Alon

Bibliography I



Robert Andrews and Michael A. Forbes.

Ideals, determinants, and straightening: Proving and using lower bounds for polynomial ideals.

In *STOC '22: 54th Annual ACM SIGACT Symposium on Theory of Computing, Rome, Italy, June 20 - 24, 2022*, pages 389–402. ACM, 2022.



Winfried Bruns and Aldo Conca.

Gröbner bases and determinantal ideals.

In *Commutative Algebra, Singularities and Computer Algebra*, pages 9–66. Springer Netherlands, 2003.

Bibliography II



Peter Bürgisser.

Completeness and Reduction in Algebraic Complexity Theory.

Springer Berlin Heidelberg, 2000.



Peter Bürgisser.

The complexity of factors of multivariate polynomials.

Foundations of Computational Mathematics, 4(4):369–396, September 2004.

Bibliography III



Aldo Conca.

Straightening Law and Powers of Determinantal Ideals of Hankel Matrices.

Advances in Mathematics, 138(2):263–292, March 1998.



Corrado de Concini, David Eisenbud, and Claudio Procesi.

Young diagrams and determinantal varieties.

Inventiones mathematicae, 56(2):129–165, 1980.



Corrado de Concini, David Eisenbud, and Claudio Procesi.

Hodge Algebras, volume 91 of *Astérisque*.

Société mathématique de France, 1982.

Bibliography IV



Anakin Dey and Zeyu Guo.

Debordering closure results in determinantal and pfaffian ideals.

ITCS, 2025.



Pranjal Dutta and Vladimir Lysikov.

Recent Advances in Debordering Methods, 2025.



Peter Doubilet, Gian-Carlo Rota, and Joel Stein.

On the foundations of combinatorial theory: Ix combinatorial methods in invariant theory.

Studies in Applied Mathematics, 53(3):185–216, 1974.

Bibliography V



David Eisenbud.

Introduction to algebras with straightening laws.

In *Ring Theory and Algebra III*, volume 55, pages 243–268. Marcel Dekker, Inc., 1980.



Joshua A. Grochow.

Complexity in ideals of polynomials: Questions on algebraic complexity of circuits and proofs.

Bulleting of the EATCS, 131, 2020.

Bibliography VI



Adam R. Klivans and Daniel Spielman.

Randomness efficient identity testing of multivariate polynomials.

In Proceedings of the 33rd Annual ACM Symposium on Theory of Computing, pages 216–223, 2001.



Rufus Oldenburger.

Polynomials in Several Variables.

41(3):694–710, 1940.

Bibliography VII



A. Ostrowski.

On Two Problems in Abstract Algebra Connected with Horner's Rule.

In *Studies in Mathematics and Mechanics*, pages 40–48. Academic Press, 1954.



Pan.

Methods of computing values of polynomials.

21(1):105–136, 1966.